Bit / Kernel Sensitivity Table (IV-B Local Reproduction)

Source: logs/iv_b_mb256/seed_42/bit_{10,13,14}/iv_b_summary_raw.csv and matching checkpoints metrics.jsonl. Baseline final eval loss = 4.317315.

Final evaluation loss delta vs baseline

cell text: final_loss - baseline; NaN = final evaluation produced NaN

	Forward kernels				Backward kernels								
	FP1	FP2	FP3	FP4	BP1	BP2	BP3	BP4	BP5	BP6	BP7	BP8	BP9
bit 10	+0.004	-0.001	-0.000	+0.024	+0.003	-0.001	+0.000	+0.065	+0.002	+0.001	-0.001	+0.011	+0.009
bit 13	+0.001	-0.000	+0.001	-0.001	-0.000	+0.001	+0.001	+0.000	+0.062	+0.152	NaN	+0.372	+0.288
bit 14	NaN	NaN	NaN	NaN	+0.061	+0.061	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Final parameter difference

cell text: L2 distance to paired baseline checkpoint

	Forward kernels				Backward kernels								
	FP1	FP2	FP3	FP4	BP1	BP2	BP3	BP4	BP5	BP6	BP7	BP8	BP9
bit 10	92	67	50	83	74	58	64	115	59	71	63	78	80
bit 13	56	52	62	65	49	52	55	55	96	149	NaN	151	156
bit 14	NaN	NaN	NaN	NaN	95	95	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Maximum gradient norm before clipping

cell text: max over training span; Inf marks non-finite gradient norm observed

	Forward kernels				Backward kernels								
	FP1	FP2	FP3	FP4	BP1	BP2	BP3	BP4	BP5	BP6	BP7	BP8	BP9
bit 10	2.3	2.7	2.3	2.3	2.3	2.8	3.1	1.8e+08	2.3	2.3	2.4	2.3	3
bit 13	2.3	2.3	2.3	2.3	2.3	2.3	2.3	3.2	2.4e+16	223	Inf	Inf	4.4e+04
bit 14	Inf	Inf	Inf	Inf	Inf	Inf	Inf	Inf	Inf	Inf	Inf	Inf	Inf

Maximum attention logits

cell text: max over training span; Inf marks non-finite attention logits observed

	Forward kernels				Backward kernels								
	FP1	FP2	FP3	FP4	BP1	BP2	BP3	BP4	BP5	BP6	BP7	BP8	BP9
bit 10	540	3920	40	45	56	50	52	54	49	52	47	44	52
bit 13	44	2.6e+20	57	53	42	53	47	48	49	46	Inf	50	97
bit 14	Inf	Inf	Inf	Inf	44	44	Inf	Inf	Inf	Inf	Inf	Inf	Inf

Notes: FP = forward-pass profiled GEMM kernel; BP = backward-pass profiled GEMM kernel. Bit 14 is a strong saturation case; NaN/Inf cells should not be interpreted as fine-grained kernel ranking.